Audio Output Obtained from Different Diodes — Table 1

Diode number	Description	Forward voltage Vf (mV)	Output without inductor (mV), 18m antenna	Output with an inductor (mV), 18m antenna
No number, Sold as 1N34A	Glass, Red Band Germanium	240	85	125
No number Sold as 1N34A	Glass, two Black bands	260	65	110
No number Sold as 1N34A	Glass, one Black Band	570	30	40
No number Sold as 1N34A	Glass, from China, Orange body/Black band	270	75	110
CV7127	Glass, Germanium	225	109	136
A9K (Russian)	Glass, Germanium	244	90	140
1N60P	Glass, Orange body	240	109	125
1N914	Glass, Silicon	600	30	35
1N270	Orange body	260	103	90
1N5817/5818	Schottky	160	100	140

diode. The failure to receive local radio stations (despite long antenna and good earth) is mostly due to the diode. One has to check the forward voltage as mentioned earlier while using the diode and try several diodes for getting good results. We used Schottky diode 1N5817/5818 instead of germanium diode and successfully received radio programmes with an external antenna longer than 18 metres.

Table 1 shows the audio output obtained from different diodes. The output of the diode is connected to a digital multimeter (Mastech 830L) in the place of headphone and measured as DC millivolts. This data is pertaining to Chennai-A radio station (720kHz) with a power of 200 kilowatts. The crystal radio is located about 45 kilometres away from the transmitter. Antenna length is 18 metres of 20SWG enameled copper wire. The audio output obtained from different diodes may vary based on the height and length of the external antenna and also depending on the distance from the transmitter.

Caution. There is a high chance of electrical shock if electrical earth point is not identified and handled properly. Novice experimenters are advised to consult an experienced electrical engineer before using an earth connection from wall socket. EFY

Dr N. Mohan retired as assistant professor and in-charge of instrumentation lab, in the dept. of botany, University of Madras. He is a radio and electronics hobbyist for more than 40 years.

Prof. S. Ananthi recently retired as a professor and head of the department of Network Systems and Information Technology, University of Madras.

K. Padmanabhan retired as a professor at A.C. Tech and University of Madras.